

The interaction between rotational pore topology, Reynolds-number-dependent instability, and coherent flow organization in structured packed porous systems remains insufficiently understood.

1. Introduction

1.1 Packed porous media and transport limitations

Purpose

- Importance of mixing and transport in packed beds
 - Need for pore-scale understanding
 - Nonuniformity and anisotropic transport
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1.2 Limitations of conventional porous-media models

Purpose

- Darcy/Ergun limitations
- Loss of local flow structures in averaged models
- Importance of coherent structures

Data Needed

- None
-

1.3 Structured rotational porous geometries

Purpose

- Explain rotational topology concept
- Periodicity and deterministic flow organization
- Difference between regular and irregular arrangements

Data Needed

- Geometry CAD/images
 - Rotation definitions (30°, 40°, irregular)
-

1.4 Literature gap and objectives

Purpose

- Lack of understanding of:
 - topology-induced flow organization
 - instability transition
 - effect of irregularity
- Define objectives of validation + physics study

Data Needed

- None
-

2. Experimental and Numerical Methodology

2.1 Experimental setup

Purpose

- Describe modular test section
- PIV measurements
- Optical access
- Reynolds number range

Data Needed

- Experimental setup image
 - Measurement plane definitions
 - Flow conditions
 - PIV parameters
-

2.2 Geometry/topology definition

Purpose

- Define:
 - rotational increment
 - periodicity
 - alignment length
 - irregular configuration

Data Needed

- Geometry images
- CAD slices
- Unit-cell illustrations

2.3 LBM numerical methodology

Purpose

- D3Q19/D3Q27
- collision model
- boundary conditions
- forcing scheme

Data Needed

- Simulation parameters
 - Lattice resolution
 - Time step
 - Relaxation parameters
-

2.4 Numerical convergence studies

Purpose

- Grid independence
- Time independence

Data Needed

- Velocity profiles for multiple grids
 - Error convergence plots
-

2.5 Experimental-to-numerical comparison methodology

Purpose

- Projection methodology
- Plane extraction
- Coordinate transformation
- Averaging procedure

Data Needed

- Experimental planes
 - Numerical plane extraction method
 - Averaging definitions
-

3. Validation Strategy

3.1 Global validation

Show

- Mean velocity
- Maximum velocity
- Flow rate consistency
- Density-based pressure-drop surrogate

Data Needed

- Mean inlet/outlet velocity
 - Density difference ($p_{in} - p_{out}$)
 - Flow rate values
-

3.2 Local validation

Show

- Velocity profiles
- Jet locations
- Recirculation regions
- Velocity contours

Data Needed

- PIV velocity fields
 - LBM velocity fields
 - Representative cross-sections
-

3.3 Statistical validation

Show

- RMSE
- Normalized error
- Pearson correlation
- TKE comparison
- Fluctuation intensity comparison

Data Needed

- Experimental statistics
- Numerical statistics
- Time-resolved data

3.4 Structural/topological validation

Show

- Streamline organization
- Coherent jet paths
- Vortex structure agreement
- Recirculation topology

Data Needed

- Streamlines
 - Instantaneous flow fields
 - Vortex identification fields
-

4. Results and Discussion

4.1 Flow organization in periodic rotational geometries

Show

- Jet formation
- Streamline locking
- Coherent transport channels
- Recirculation structures

Compare

- 30° vs 40°

Data Needed

- Mean velocity contours
 - Streamlines
 - Cross-sectional flow fields
-

4.2 Reynolds-number-dependent transition

Show

- $Re_{100} \rightarrow$ stable
- $Re_{300} \rightarrow$ transitional
- $Re_{500} \rightarrow$ fluctuating

Analyze

- Symmetry loss
- Jet instability
- Fluctuation growth
- Vortex interaction

Data Needed

- Instantaneous snapshots
 - Time-resolved velocity data
 - Fluctuation maps
 - TKE maps
-

4.3 Temporal and fluctuation analysis

Show

- Velocity probe signals
- FFT spectra
- Oscillation frequencies
- Dominant flow modes

Data Needed

- Probe time series
 - FFT analysis
 - Frequency spectra
-

4.4 Effect of topological irregularity

Show

- Broken symmetry
- Jet redistribution
- Altered recirculation
- Enhanced transverse transport

Data Needed

- Irregular geometry results
 - Comparison with regular case
 - Streamline fields
-

4.5 Coherent structure and vortex analysis

Show

- Q-criterion
- λ^2 criterion
- Swirling strength
- Coherent vortical regions

Data Needed

- Velocity-gradient tensor fields
 - Vortex identification metrics
-

4.6 Implications for transport and reactor design

Discuss

- Mixing enhancement
- Residence-time implications
- Flow homogenization
- Reactive transport relevance

Data Needed

- Derived global metrics
 - Transport interpretation
-

5. Conclusions

Main conclusions

- Rotational topology organizes coherent flow structures
- Reynolds number controls instability transition
- Irregularity disrupts periodic flow locking
- Validated LBM captures pore-scale dynamics
- Structured topology can guide reactor design

Data Needed

- None (summary section)
-

Essential Figures

Figure 1

Geometry and topology definition

Figure 2

Experimental + numerical setup

Figure 3

Validation metrics overview

Figure 4

Mean flow organization (30°)

Figure 5

Mean flow organization (40°)

Figure 6

Instantaneous structures and fluctuations

Figure 7

Reynolds-number transition

Figure 8

Irregular topology effects

Figure 9

Vortex/coherent structure analysis

Figure 10

Global metrics summary

Minimum Required Data Extraction Checklist

- Mean velocity fields
- Instantaneous velocity fields
- Streamlines
- Velocity profiles

- Density difference (pressure surrogate)
- TKE maps
- Velocity fluctuations
- FFT/probe signals
- Vortex metrics
- Grid independence results
- Time independence results
- Global flow metrics